Expert Session

An informal expert session on pricing issues

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Modifications of insurances

New insurances and their modifications

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What are we talking about?

- OECD defines how to price new transactions:
  - Minimum pricing methodologies for OECD transactions: MPR, MB
  - ECA-specific pricing methodologies for all other transactions
  - Risk premium is 80% of total premium

However:

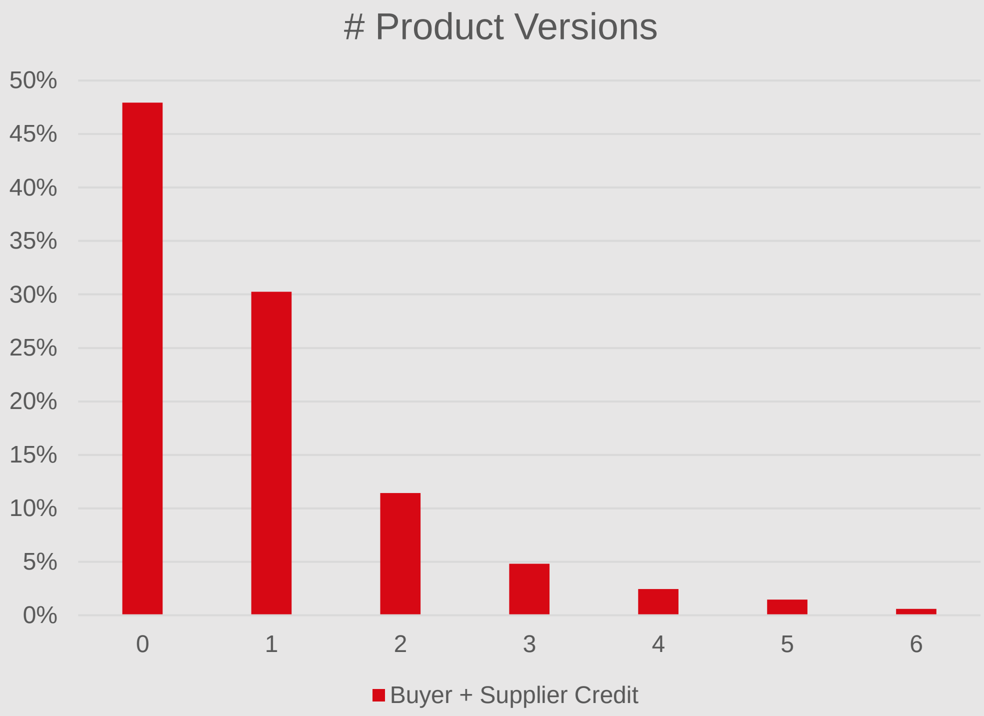
- OECD does not help on pricing modifications of insurances
- ECA-specific pricing methodologies for all changes

Why is this relevant?

Modifications are frequent: Less than half of the insured buyer + supplier credits stay unmodified.

Why?

- Delivery contract may contain some flexibility in terms of dates (delivery)
- Definition of SPOC might be flexible (commissioning date)
- Delivery is delayed
- Repayments are anticipated
- Repayments are delayed (restructuring)
- Whole transaction is repaid early.



SERV Experience with pricing modifications of insurances

Situation at SERV before 2021:

- No SERV methodology for pricing modifications.
- Modifications were priced as difference to corresponding new transaction.
- With early repayments, administrative premium was reimbursed → net loss

Approach:

- In 2021, SERV started to apply a new calculation method for (full or partial) early repayments or reductions of amounts.
- Since July 2023, SERV applies the same methodology to all modifications.

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Graphical Examples

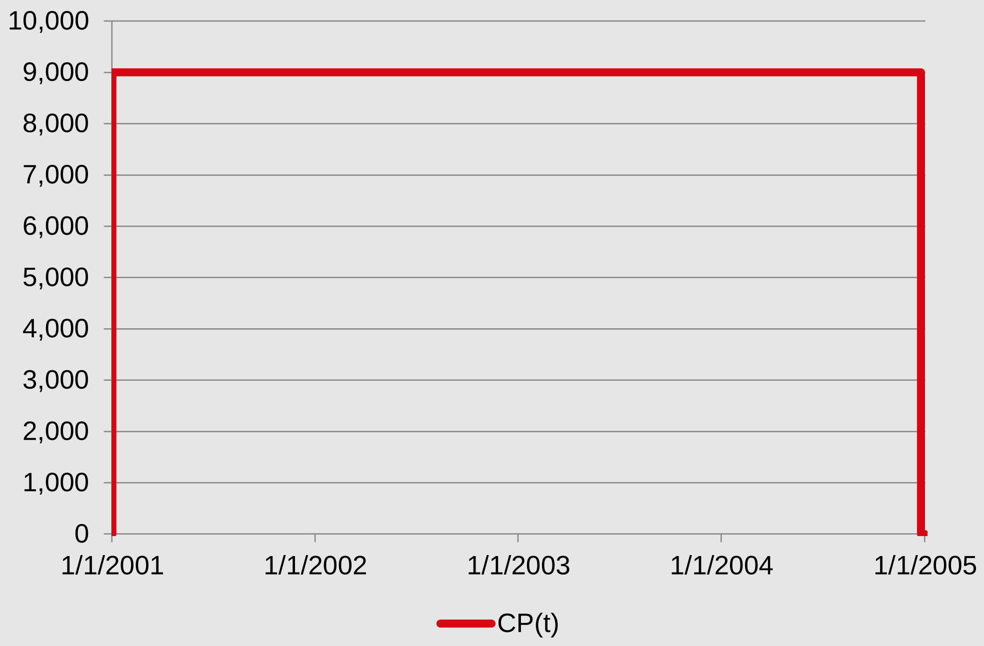
Two graphical examples for calculation

Risk represented by area: Example 1

To show our approach, we use a graphical approach.

As an example, we take the following transaction:

- Buyer Credit
- Cover: 9’000 EUR (might be 9’474 EUR with 95% cover)
- Credit period: 4 years
- Total premium: 2’250 EUR → Risk premium: 1’800 EUR



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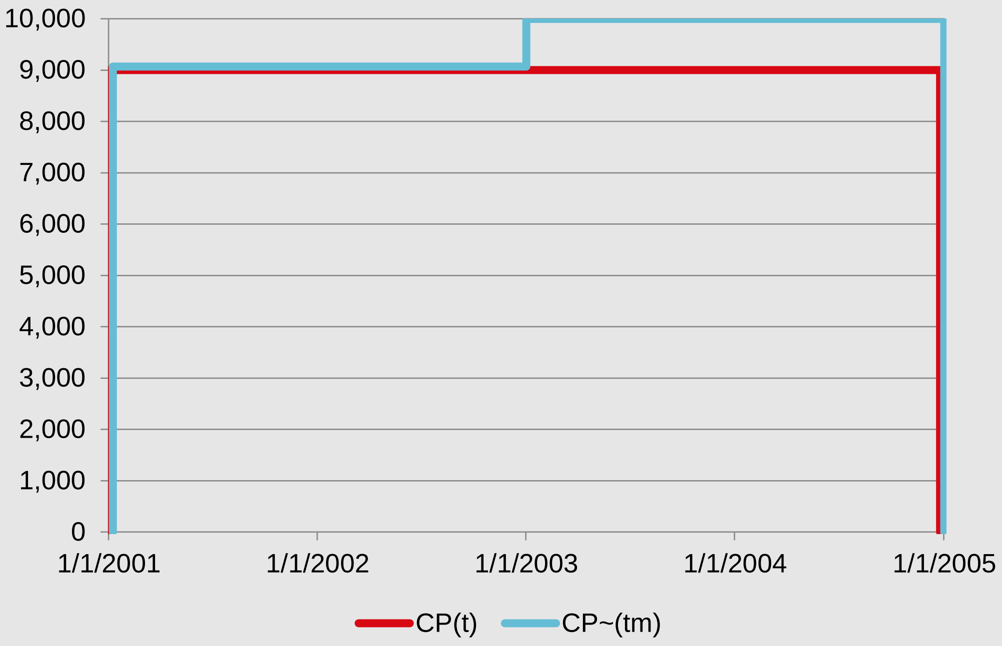
Risk represented by area: Example 1

As of 1.1.2003, an increase of amount is applied for:

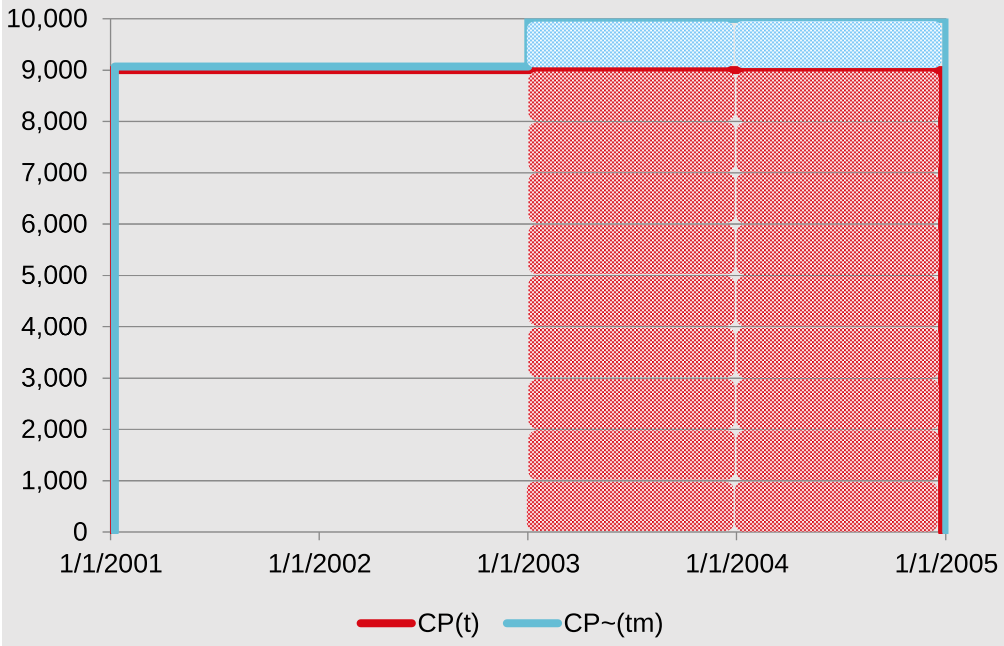
- New Cover: 10’000 EUR
- Date of change: 1.1.2023



Risk represented by area: Example 1

How does the area increase?

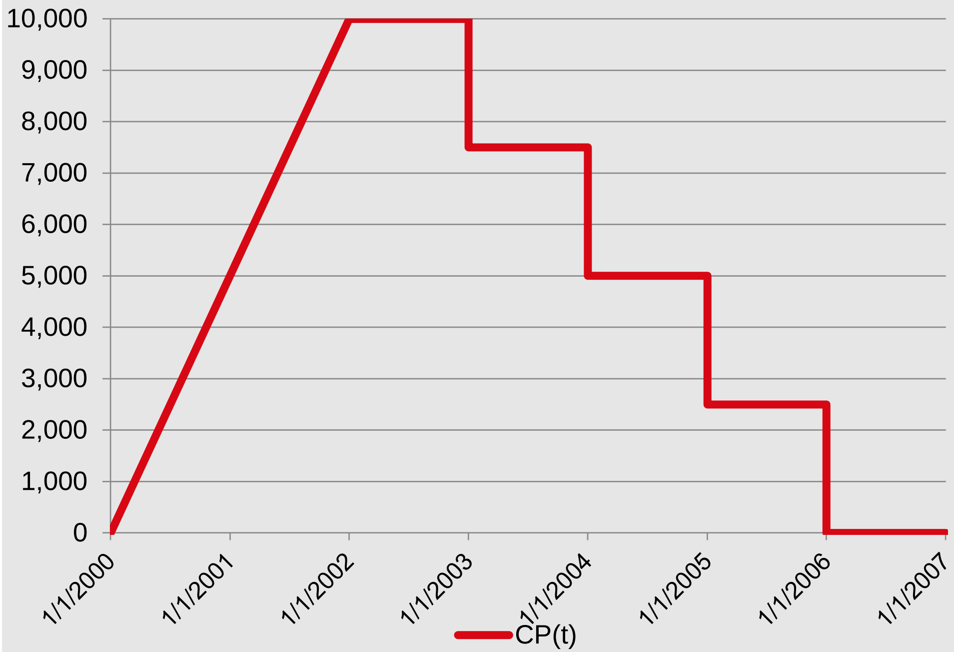
- Red area: 18 «units»,
  corresponds to unearned premium of EUR 900
- Blue area: 2 «units»
- ➔ Increase of Area after 1.1.2003: 2/18=11.1%
- ➔ Increase of Risk Premium: 11.1% x EUR 900 = EUR 100
- ➔ Increase of Total Premium: EUR 100 / 0.8 = EUR 125



Risk represented by area: Example 2

Typical structure of Buyer Credit:

- Amortizing Buyer Credit
- Cover: 10’000 EUR (might be 10’526 EUR with 95% cover)
- Disbursement Period: 2 years
- Credit period: 4 years
- Total premium: 2’187 EUR → Risk premium: 1’750 EUR



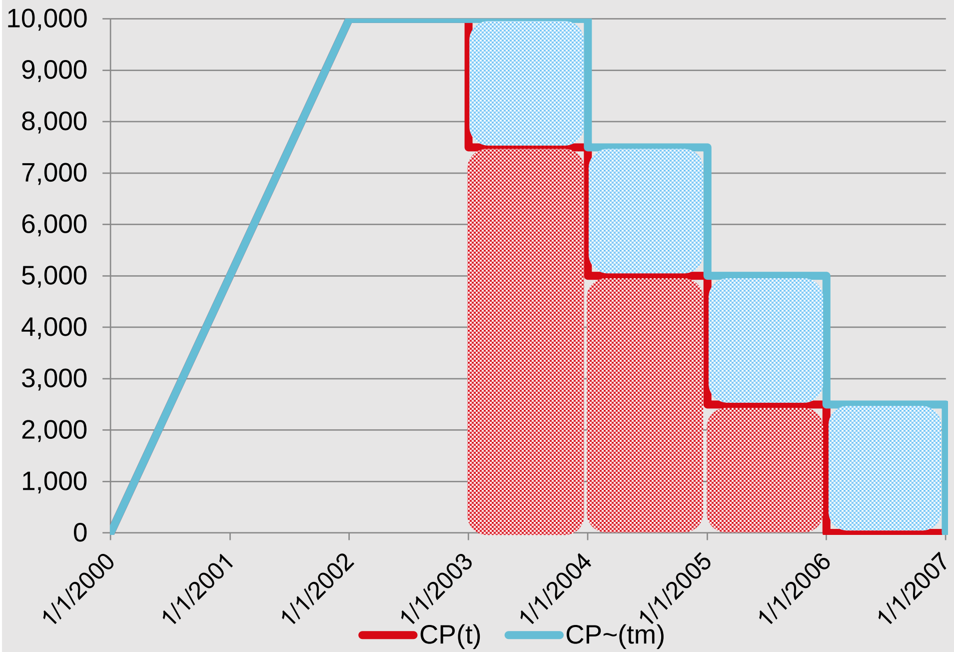
Risk represented by area: Example 2

As of 1.1.2003, an change is applied for:

- Shift of one amortization by one year
- Date of change: 1.1.2023
- Unearned premium: EUR 750

How does the area increase?

- Red area: 6 «units»
- Blue area: 4 «units»
- ➔ Increase of Area after 1.1.2003: 4/6=66.7%
- ➔ Increase of Risk Premium: 66.7% x EUR 750 = EUR 500
- ➔ Increase of Total Premium: EUR 500 / 0.8 = EUR 625



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Formal Approach

Establishing a mathematical framework for the pricing of modifications

Specific Risk Premium as Premium Density

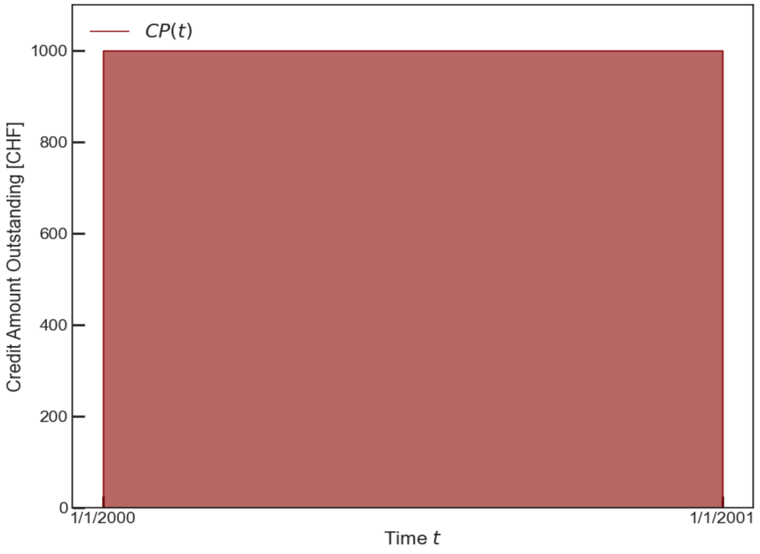
Introduction of the so-called *specific risk premium:*

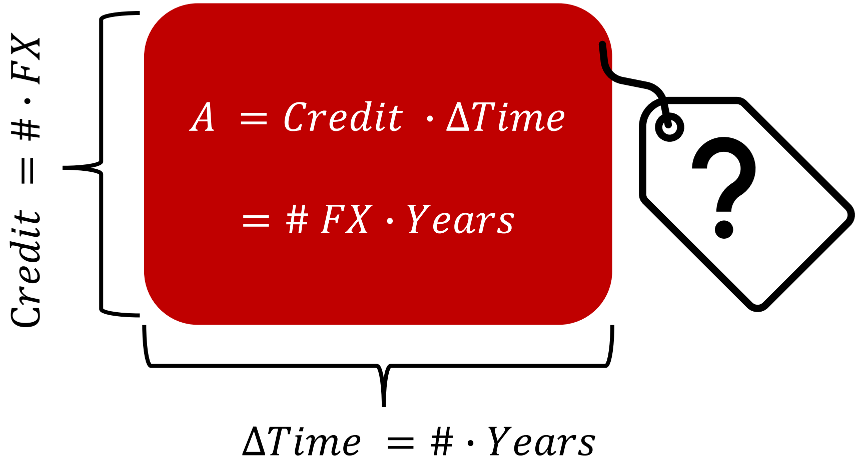
$$SRP = \frac{RP}{A} = \frac{RP}{\int_{t=-\infty}^{\infty} CP(t) \cdot dt}$$

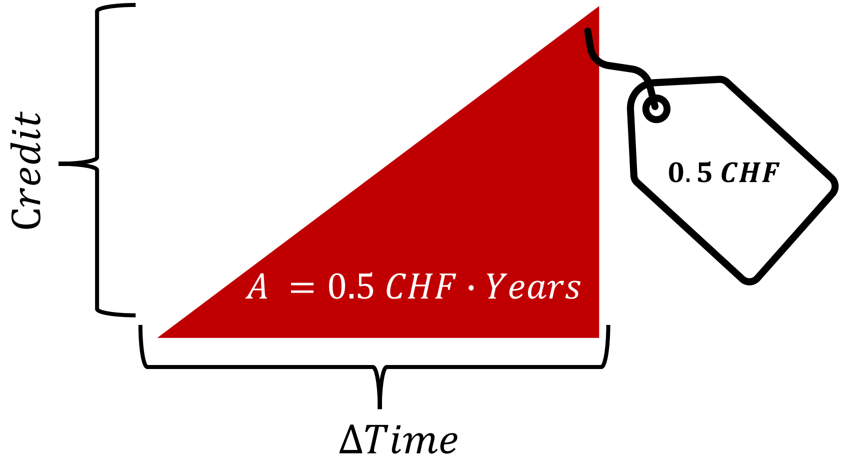
- RP – risk premium, A – risk area under cover profile $CP(t)$
- ➢ SRP quantifies the risk premium due for a risk area of unity

Example

- $RP = 1\,000\ CHF$
- $A = 1\,000\ CHF \cdot Years$
- ➢ $SRP = 1.0 \cdot \frac{1}{Years}$







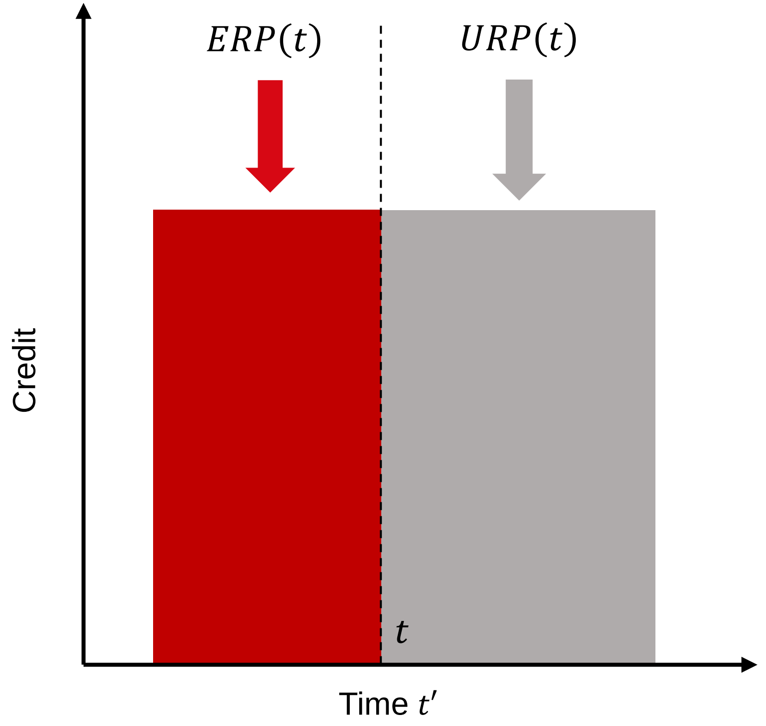
Earned and Unearned Premiums

- The earned risk premium $ERP(t)$ at time t can be defined via the SRP as follows

$$ERP(t) = SRP \cdot \int_{t'=-\infty}^{t} CP(t') \cdot dt' .$$

- The unearned risk premium $URP(t)$ at time t is the difference of the total risk premium and the amount already earned thereof until time t

$$URP(t) = RP - ERP(t) = SRP \cdot \int_{t'=t}^{\infty} CP(t') \cdot dt' .$$



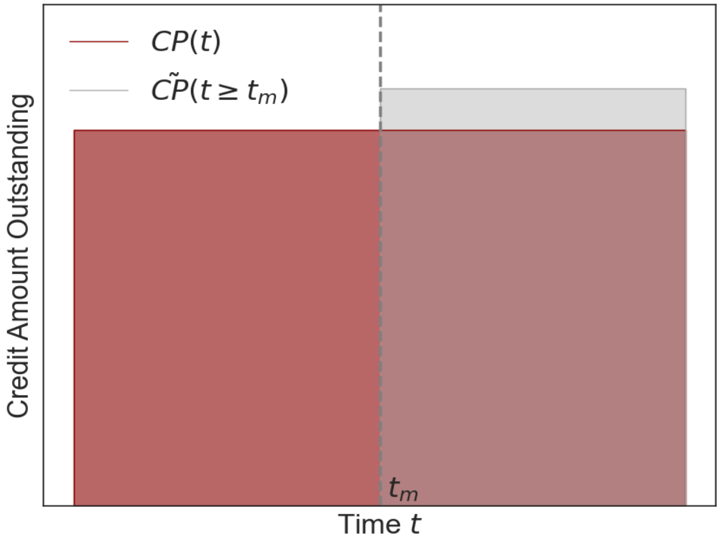
The earned and unearned risk premiums are related to the time-sliced cover profile via the specific risk premium and ensure consistency between premium calculation and accrual.

Pricing Framework for Modifications

- $\widetilde{CP}(t)$ and $CP(t)$ are the new and old cover profiles corresponding to the initial and modified transaction, respectively
- Let $\tilde{A}(t \geq t_m) = \int_{t=t_m}^{\infty} \widetilde{CP}(t) \cdot dt$ and $A(t \geq t_m) = \int_{t=t_m}^{\infty} CP(t) \cdot dt$
- The change in risk premium is the difference in unearned risk premiums counting from the date of modification t_m

$$\Delta RP = \widetilde{RP} - RP = \int_{t=-\infty}^{\infty} SRP \cdot \left[\widetilde{CP}(t) - CP(t)\right] \cdot dt \ = SRP \cdot \int_{t=t_m}^{\infty} \left[\widetilde{CP}(t) - CP(t)\right] \cdot dt$$

$$= \widetilde{URP}(t_m) - URP(t_m) = SRP \cdot \left[\tilde{A}(t \geq t_m) - A(t \geq t_m)\right]$$



Risk premium — $\Delta RP = SRP \cdot \Delta A(t_m)$ — Cover profile topography

Constant* from initial pricing

*
- Generally, $SRP(t) \neq \widetilde{SRP}(t)$.
- In absence of reevaluations (e.g. letter rating updates):

$$SRP(t) = SRP = \widetilde{SRP}$$

Approach only requires knowledge of the cover profile topography and is therefore universally applicable across different export cases, policy types, etc.

Example I: Increased Risk Coverage

- At $t_m = 1/1/2002$ the covered amount increases from 9000 to 10000 CHF

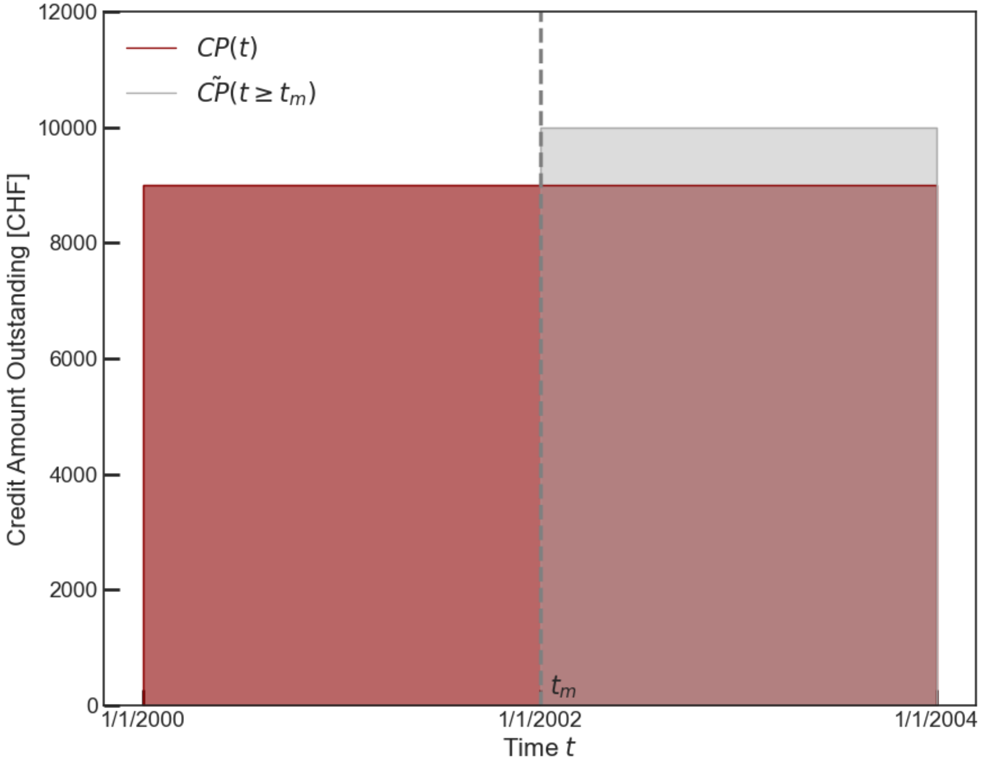
- The specific risk premium is $SRP = 0.05 \cdot \frac{1}{Years}$

- Additional risk premium due to the modification reads:

$$\begin{aligned}\Delta RP &= SRP \cdot \left[\tilde{A}(t_m) - A(t_m)\right] \\ &= 0.05 \cdot (2 \cdot 10000 - 2 \cdot 9000)\ \mathrm{CHF} \\ &= 100\ \mathrm{CHF}\end{aligned}$$

- The total and administrative premium are*

$$\Delta P = \frac{\Delta RP}{\omega} = \frac{100.0}{0.8} = 125\ \mathrm{CHF}, \qquad \Delta AP = \Delta P - \Delta RP = 25\ \mathrm{CHF}$$



*$\omega = 0.8$ – percentage of risk premium from total premium

Example II: Extension of Repayment Period

- Deferral of all time-periodic repayments of amount 2500 CHF by one repayment time interval of $1\ \mathrm{Year}$

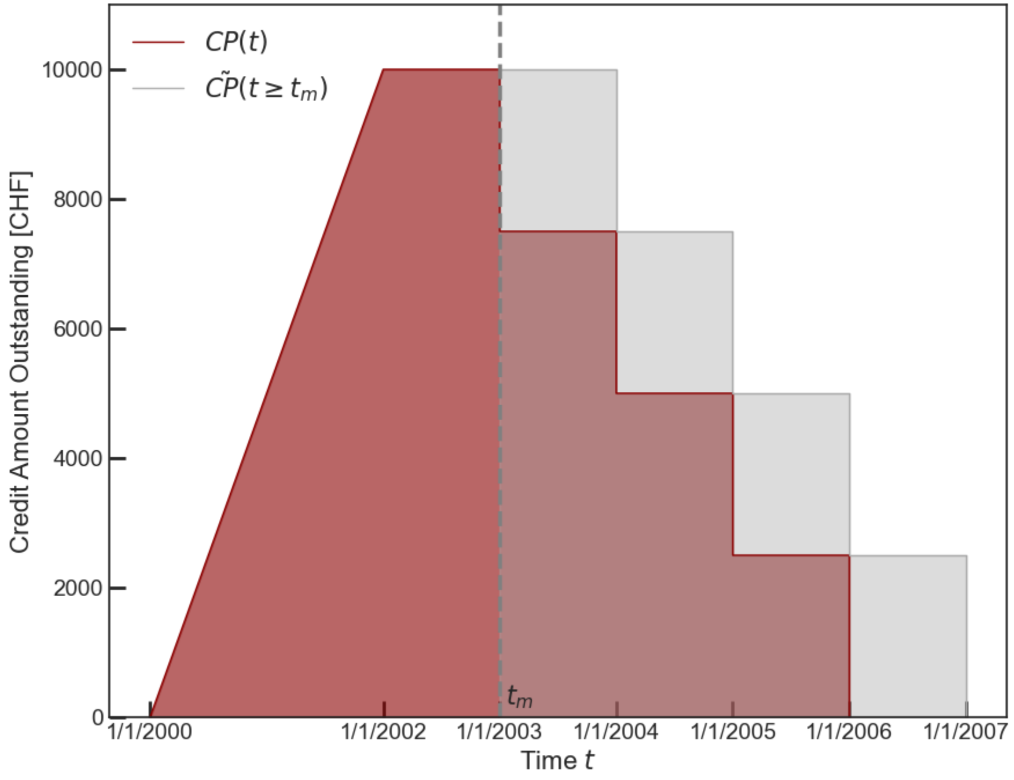
- The specific risk premium is $SRP = 0.05 \cdot \frac{1}{Years}$

- Additional risk premium due to the modification reads:

$$\begin{aligned}\Delta RP &= SRP \cdot \left[\tilde{A}(t_m) - A(t_m)\right] \\ &= 0.05 \cdot (25000 - 15000)\ \mathrm{CHF} \\ &= 500\ \mathrm{CHF}\end{aligned}$$

- The total and administrative premium are

$$\Delta P = \frac{\Delta RP}{\omega} = \frac{500.0}{0.8} = 625\ \mathrm{CHF}, \qquad \Delta AP = \Delta P - \Delta RP = 125\ \mathrm{CHF}$$



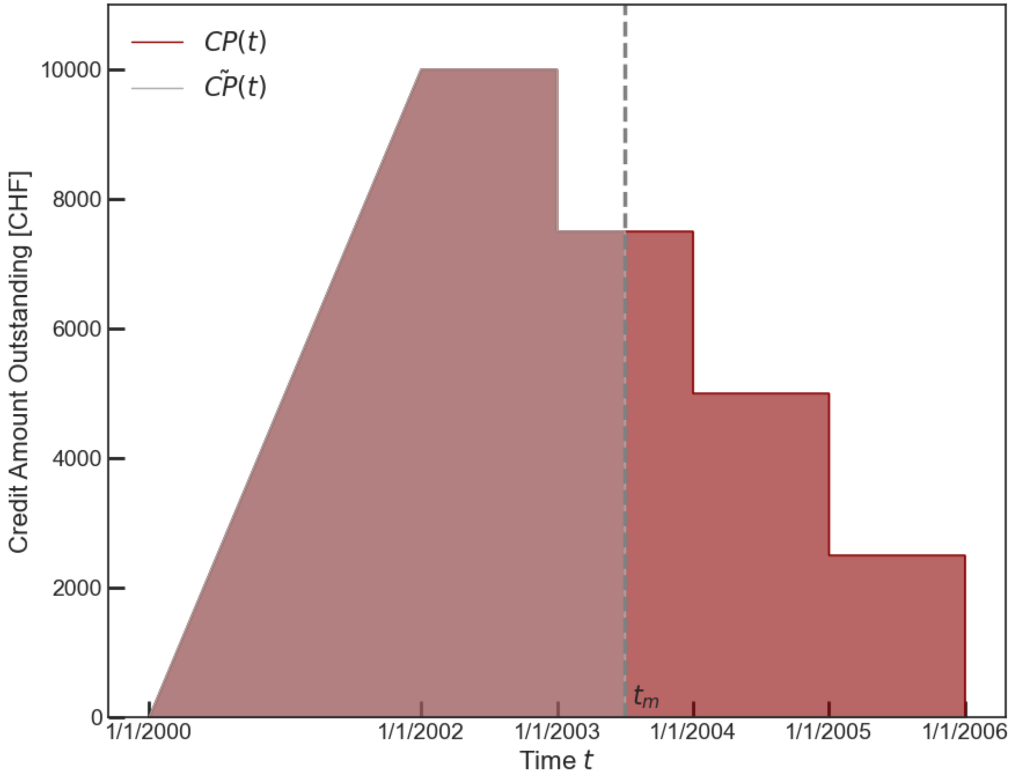
Example III: Premature Termination

- Termination of an active insurance at time t_m
- The specific risk premium is $SRP = 0.05 \cdot \frac{1}{Years}$
- Risk premium *refund* due to the modification reads:

$$\begin{aligned} \Delta RP &= SRP \cdot \left[\tilde{A}(t_m) - A(t_m)\right] \\ &= 0.05 \cdot (0 - 11250)\ \mathrm{CHF} \\ &= -562.5\ \mathrm{CHF} \end{aligned}$$

- The total premium and administrative deduction are*

$$\Delta P = \Omega \cdot \Delta RP = -450\ \mathrm{CHF}, \qquad AD = \Delta P - \Delta RP = 112.5\ \mathrm{CHF}$$



* $\Omega = 0.8$ – percentage of risk premium refund from total premium refund

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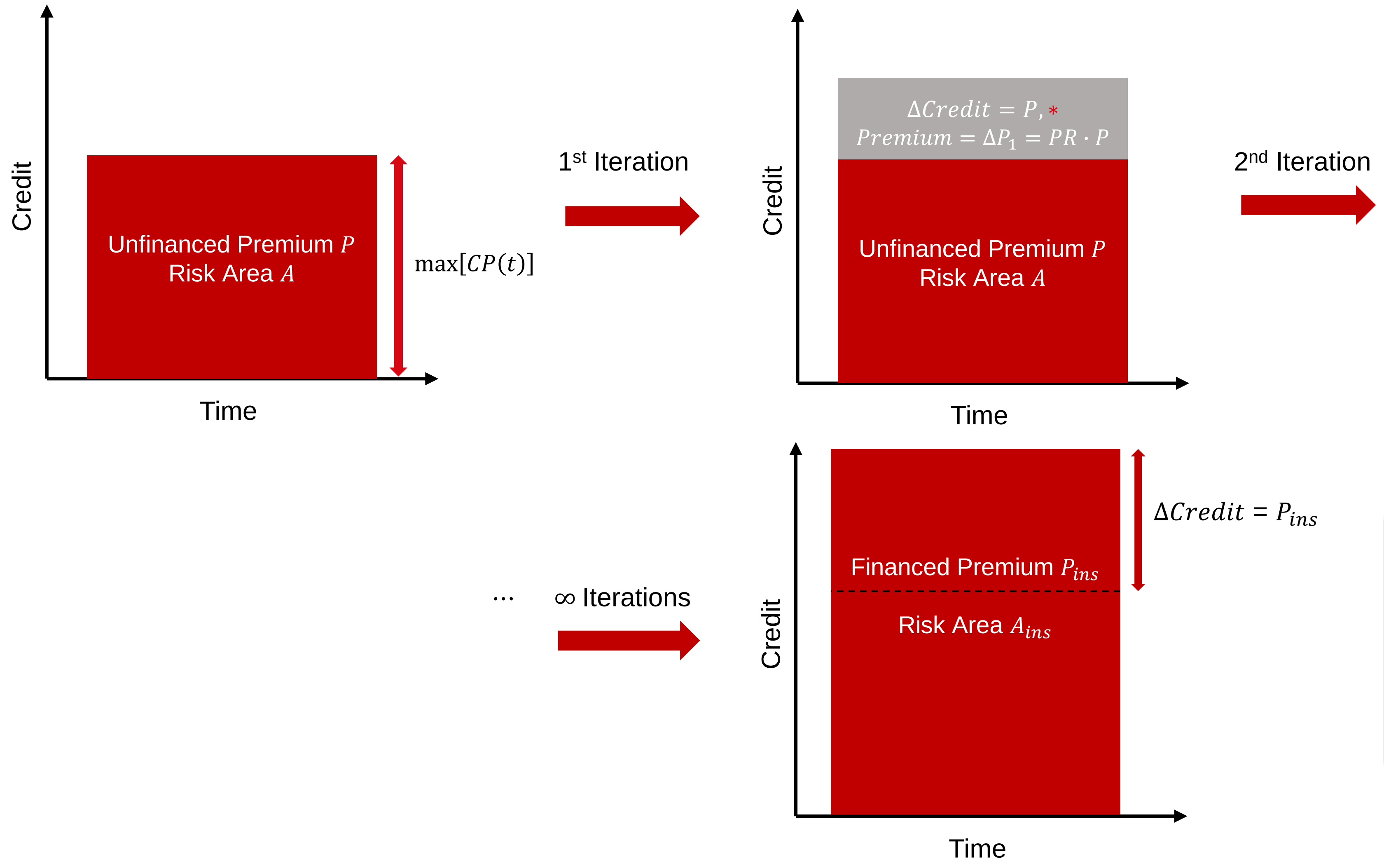
Premium Financing

Development of a self-consistent solution for the premium financing of modifications

Recap: Premium Financing



The iterations correspond to a convergent geometrical series:

$$P_{ins} = P + PR \cdot P + PR^2 \cdot P + \cdots =$$

$$= \sum_{n=0}^{\infty} PR^n \cdot P = \frac{1}{1 - PR} \cdot P \geq P$$

*$PR = \frac{P}{\max[CP(t)]}$ –$(unfinanced)$ $premium\ rate$

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Premium Financing of Modifications

- Financing of a premium surcharge $\Delta P > 0$ corresponds to an increased coverage $\tilde{A}(t \geq t_m)_{ins}$ at the expense of a higher premium $\Delta P_{ins} \geq \Delta P$
- Mathematically, there are two unknown variables:

$$\Delta P_{ins} = \kappa_{\Delta P} \cdot \Delta P \ (1), \qquad \tilde{A}_{ins}(t \geq t_m) = \kappa_{\tilde{A}(t \geq t_m)} \cdot \tilde{A}(t \geq t_m) \ (2)$$

- Business logic implies two boundary conditions:

$$SRP_{ins} = SRP \ (1), \quad \max\left[\widetilde{CP}_{ins}(t \geq t_m)\right] = \max\left[\widetilde{CP}(t \geq t_m)\right] + \Delta P_{ins} \ (2)$$

OECD results for premium financing are contained as special cases of our solution

Solution:

$$\Delta P_{ins} = \kappa_{\Delta P} \cdot \Delta P = \frac{1}{1 - \frac{SRP \cdot \tilde{A}(t_m)}{\omega \cdot \max\left[\widetilde{CP}(t_m)\right]}} \cdot \Delta P \geq \Delta P$$

$$\tilde{A}_{ins}(t \geq t_m) = \left[1 + \kappa_{\Delta P} \cdot \frac{\Delta P}{\max\left[\widetilde{CP}(t_m)\right]}\right] \cdot \tilde{A}(t_m) \geq \tilde{A}(t_m)$$

In the limit $t_m \to -\infty$, the solutions for premium financing are recovered:

$$P_{ins} = \frac{1}{1 - PR} \cdot P \geq P,$$

$$A_{ins} = \frac{1}{1 - PR} \cdot A \geq A.$$

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Example II Revisited: Premium Financing

- Same setup as Example II:

$$SRP = 0.05 \cdot \frac{1}{\text{Years}}, \qquad \max\left[\widetilde{CP}(t_m)\right] = 10000 \text{ CHF}$$

- Results without premium financing:

$$\Delta P = 625 \text{ CHF}, \qquad \tilde{A}(t_m) = 25000 \text{ CHF} \cdot \text{Years},$$

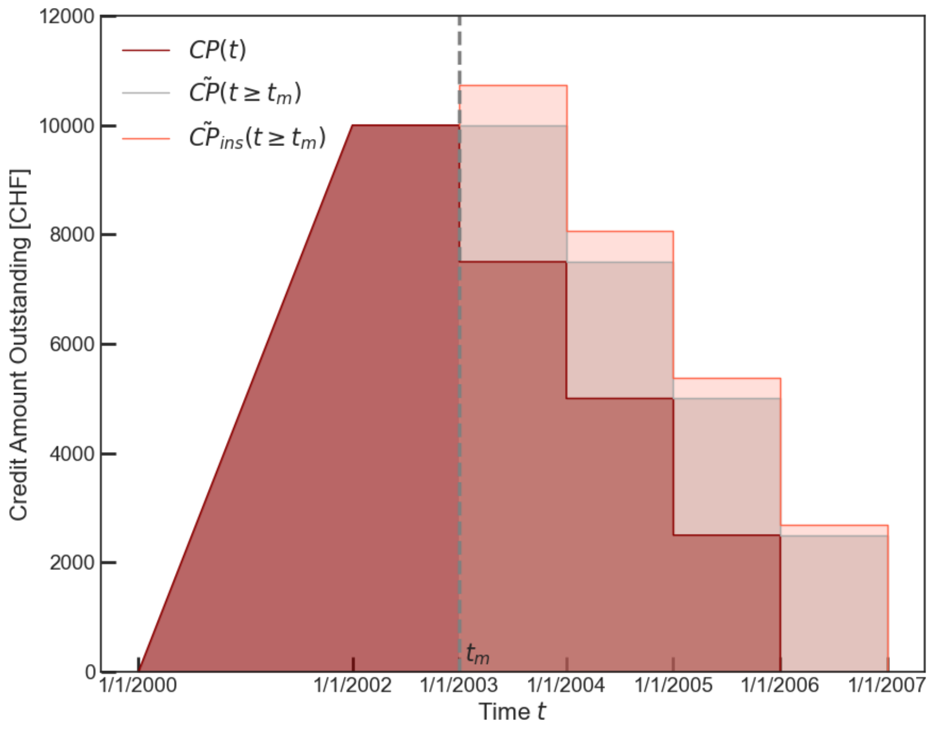
- Results for premium financing:

$$\kappa_{\Delta P} = \frac{1}{1 - \frac{\mu \cdot SRP \cdot \tilde{A}(t_m)}{\omega \cdot \max\left[\widetilde{CP}(t_m)\right]}} = \frac{1}{1 - \frac{1.0 \cdot 0.05 \cdot 25000}{0.8 \cdot 10000}} = 1.19$$

$$\kappa_{\tilde{A}(t \geq t_m)} = 1 + \kappa_{\Delta P} \cdot \frac{\Delta P}{\max\left[\widetilde{CP}(t_m)\right]} = 1.07$$

- The financed total premium and the new credit amount read:

$$\Delta P_{ins} = 744 \text{ CHF}, \qquad \max\left[\widetilde{CP}_{ins}(t \geq t_m)\right] = 10744 \text{ CHF}$$



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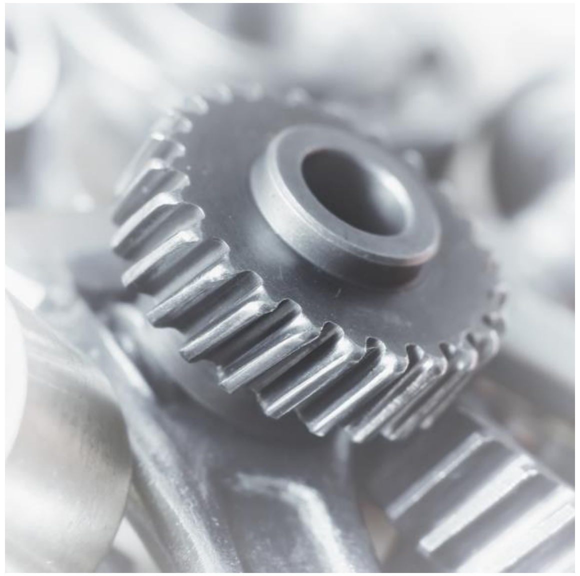
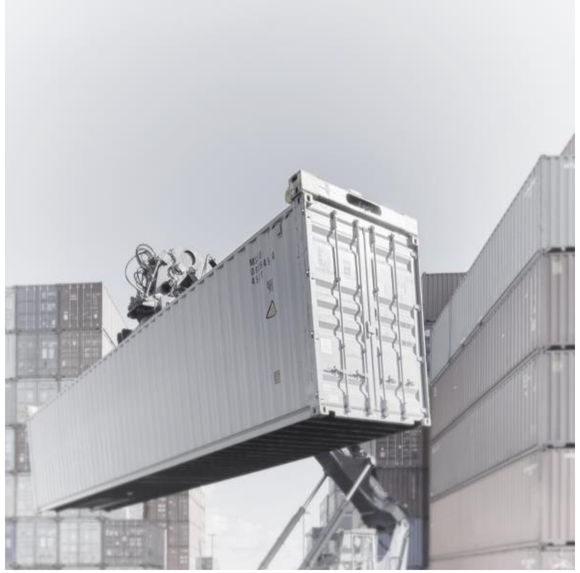


Minimise risks. Maximise exports.



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